PROBLEM 49

Particle on a Rotating Sphere Attracted to a Point

Original statement · Problems.pdf, p. 34

Relative Equilibria & Bifurcation

A point particle P of mass m slides without friction on a sphere of radius a centered at the origin of the inertial reference frame S_0 as shown in the figure. The particle is attracted by a point A in the equatorial plane, located a distance a from the surface of the sphere as shown. The force of attraction is linearly proportional [sic] the distance between this point and P, with known constant of proportionality $k > 0$. The sphere and the point A rotate rigidly about the vertical with a known angular velocity $\Omega \mathbf{k}_0$. We define the auxiliary reference frame S_1 that rotates with the sphere and point A, with $\mathbf{k}_1 = \mathbf{k}_0$ and $\mathbf{i}_1$ pointing in the direction of A. We introduce the angles ϕ and θ as shown to locate P on the sphere. Note that there is no gravity in this problem.

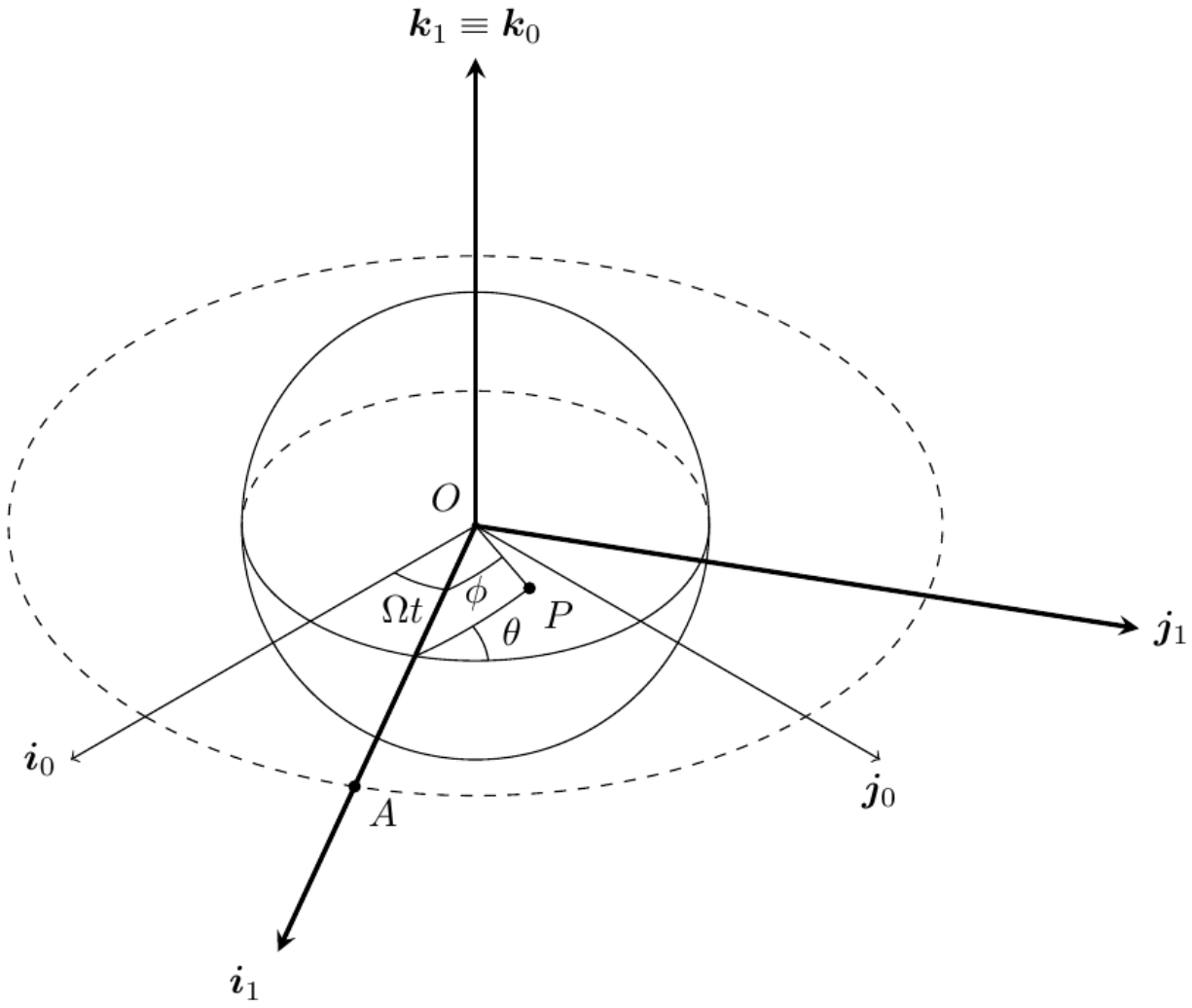


Figure: Sphere of radius a centred at O, point A on the i1 axis outside the sphere in the equatorial plane, particle P located by the angles phi and theta, frame S1 rotated by Omega t about k0 = k1 (Problems.pdf, p. 34).

(a) Express the position, velocity, and acceleration of P with respect to S_1 in terms of ϕ, θ and their derivatives. Remember to define any unit vectors that you want to use in terms of $\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0$ first.

(b) Write down the (vector) equation of motion of P in S_1, and derive from there a set of two scalar second-order differential equations for ϕ, θ that do not depend on any other unknowns. To do so, first identify and write down all the forces needed in your analysis.

(c) Is the mechanical energy of P conserved in S_1? Explain. If so, write down the corresponding conservation law.

(d)

Identify all equilibrium configurations of the system, and discuss them as a function of the ratio $m\Omega^2/k$.

(e) Assuming the functions of time $\phi(t)$ and $\theta(t)$ are known, write down the expression of the reaction of the sphere on P.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** a particle P of mass m slides on a smooth sphere of radius a centred at O. A point A of the equatorial plane, at distance a from the surface (so $OA = 2a$), attracts P with the force $k\,\overrightarrow{PA}$. Sphere and A rotate rigidly about the vertical with angular velocity $\Omega\,\mathbf{k}_0$. No gravity.
- **Hypotheses:**
  1. $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ is inertial; Ω is constant.
  2. The sphere is smooth and the contact bilateral: the reaction is radial, $\mathbf{N} = N\,\mathbf{e}_r$ with N of either sign.
  3. **Reading of the figure (angles):** ϕ is the angle at O between $\overrightarrow{OA}$ (i.e. $\mathbf{i}_1$) and $\overrightarrow{OP}$; θ is the angle, measured about the axis OA, between the plane containing $\mathbf{i}_1$ and $\overrightarrow{OP}$ and the equatorial plane (the figure draws it at the point where OA crosses the sphere, between the equator and the arc through P). So ϕ is a polar angle about the axis OA and θ the corresponding azimuth, with $\theta = 0$ in the equatorial plane on the side of $\mathbf{j}_1$ and $\theta = \pi/2$ in the upper hemisphere.
- **Frames:** $S_0 = \{O; \mathcal{B}_0\}$ inertial; $S_1 = \{O; \mathcal{B}_1\}$, $\mathcal{B}_1 = \{\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, rotating with the sphere, with $\mathbf{k}_1 = \mathbf{k}_0$ and $\mathbf{i}_1$ along $\overrightarrow{OA}$; the figure measures the angle Ωt from $\mathbf{i}_0$ to $\mathbf{i}_1$.
- **Spherical basis with polar axis OA:** $\{\mathbf{e}_r, \mathbf{e}_\phi, \mathbf{e}_\theta\}$ at P, deduced in the next section.
- **Constraints and degrees of freedom:** P has 3 coordinates and one constraint, $|\overrightarrow{OP}| = a$ (holonomic, scleronomic in S_1, smooth): CDOF $= 3 - 1 = 2$, with the angles (ϕ, θ); the unknown reaction is N along $\mathbf{e}_r$.
- **Why S_1:** in S_1 both the sphere and the attracting point A are fixed, so the attraction depends only on the position of P and the problem has no explicit time dependence; the inertia forces (Notes Eq. 4.62) take that role. With polar axis OA, the distance AP depends on ϕ only, which keeps the attraction simple.

Symbol	Meaning	SI unit
m	mass of P	kg
a	radius of the sphere ($OA = 2a$)	m
k	attraction constant	N/m
Ω	angular rate of sphere and A	rad/s
ϕ, θ	polar angle from OA, azimuth about OA	rad
N	normal reaction	N
$\Lambda = m\Omega^2/k$	ratio of centrifugal to attraction stiffness	1

Change of Basis and Position Vector

The basis $\mathcal{B}_1$ seen from above

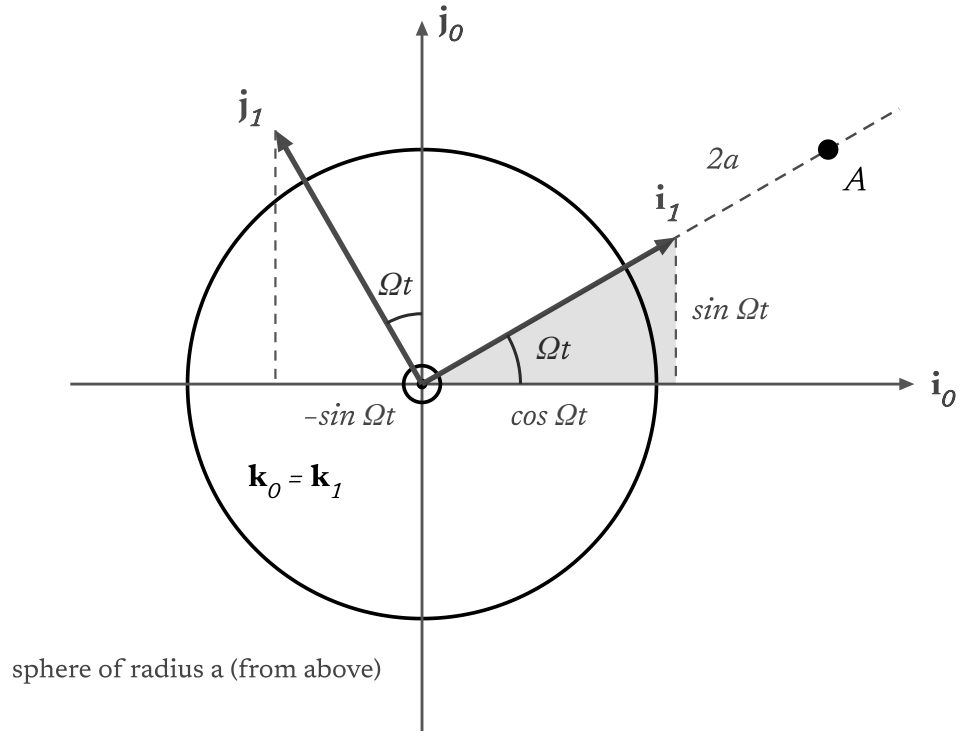


Figure 1. View from $+z$, with $\mathbf{k}_0 = \mathbf{k}_1$ out of the paper ($\odot$), so $\mathbf{j}_0$ is $\mathbf{i}_0$ turned 90° counter-clockwise. S_1 turns Ωt counter-clockwise: $\mathbf{i}_1$ has the legs $\cos\Omega t$ along $\mathbf{i}_0$ and $\sin\Omega t$ along $\mathbf{j}_0$; the tip of $\mathbf{j}_1$ moves to the side of $-\mathbf{i}_0$, hence $-\sin\Omega t$. A sits on the $\mathbf{i}_1$ axis at $OA = 2a$.

The rotation is about $\mathbf{k}_0$, so we look down from $+z$: $\mathbf{k}_0 = \mathbf{k}_1$ points out of the paper ($\odot$). With $\mathbf{i}_0$ to the right, $\mathbf{i}_0 \times \mathbf{j}_0 = \mathbf{k}_0$ requires the fingers to curl counter-clockwise from $\mathbf{i}_0$ to $\mathbf{j}_0$ for the thumb to point at the reader, so $\mathbf{j}_0$ points up the page. With $\Omega > 0$ the frame S_1 turns counter-clockwise, and after a time t:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of the shaded triangle: the leg adjacent to Ωt is $\cos \Omega t$ along $\mathbf{i}_0$ and the leg opposite to Ωt is $\sin \Omega t$ along $+\mathbf{j}_0$. So $\mathbf{i}_1 = \cos \Omega t\, \mathbf{i}_0 + \sin \Omega t\, \mathbf{j}_0$.
- $\mathbf{j}_1$ starts on $\mathbf{j}_0$ and turns the same angle counter-clockwise, so its tip moves to the side of $-\mathbf{i}_0$: leg $\cos \Omega t$ along $\mathbf{j}_0$ and leg $\sin \Omega t$ towards $-\mathbf{i}_0$. **That is the origin of the minus sign** in $\mathbf{j}_1 = -\sin \Omega t\, \mathbf{i}_0 + \cos \Omega t\, \mathbf{j}_0$.
- $\mathbf{k}_1 = \mathbf{k}_0$, the axis of the rotation.

$$\mathbf{i}_1 = \cos \Omega t\, \mathbf{i}_0 + \sin \Omega t\, \mathbf{j}_0, \qquad \mathbf{j}_1 = -\sin \Omega t\, \mathbf{i}_0 + \cos \Omega t\, \mathbf{j}_0, \qquad \mathbf{k}_1 = \mathbf{k}_0$$

$$[{}_0R_1] = \begin{pmatrix} \cos \Omega t & -\sin \Omega t & 0 \\ \sin \Omega t & \cos \Omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1 \cdot (\cos \Omega t \cos \Omega t - (-\sin \Omega t) \sin \Omega t) = \cos^2 \Omega t + \sin^2 \Omega t = 1$$

The rows are unit vectors and the first two are orthogonal ($\cos \Omega t \sin \Omega t - \sin \Omega t \cos \Omega t = 0$), so $[{}_0R_1][{}_0R_1]^T = I$.

Motion of S_1: $\mathbf{a}_0^O = \mathbf{0}$, $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{k}_1$ (simple rotation, Slide 16), $\boldsymbol{\alpha}_{10} = \mathbf{0}$. Check: by the chain rule $\frac{d\mathbf{i}_1}{dt}\big|_0 = -\Omega \sin\Omega t\,\mathbf{i}_0 + \Omega\cos\Omega t\,\mathbf{j}_0 = \Omega\,\mathbf{j}_1 = \Omega\,\mathbf{k}_1 \times \mathbf{i}_1$. The attracting point is fixed in S_1: $\mathbf{r}_1^A = 2a\,\mathbf{i}_1$ (it lies on the dashed $\mathbf{i}_1$ axis of the figure).

The azimuth θ: view along OA

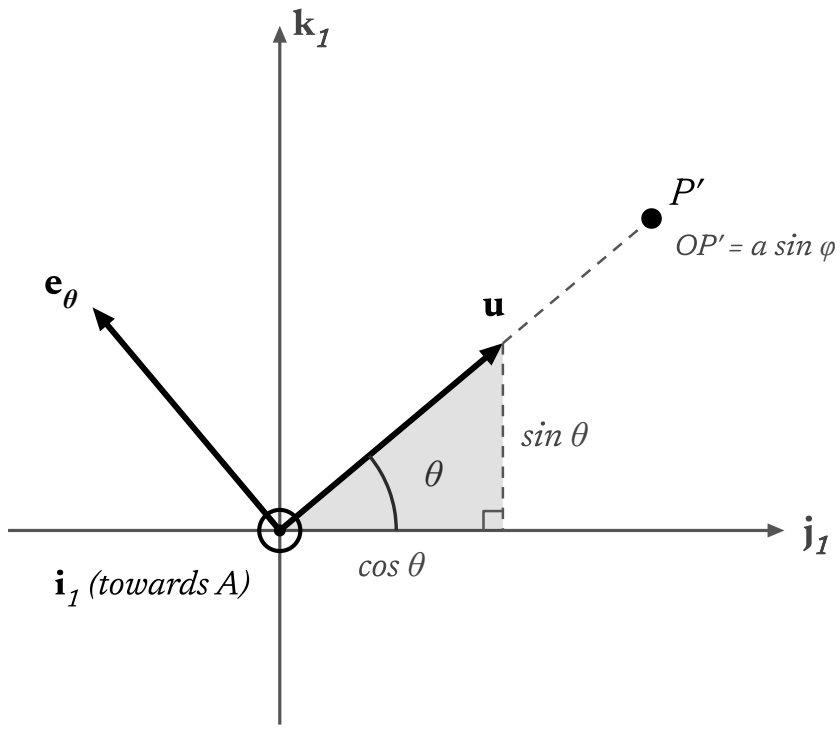


Figure 2. View along OA, with $\mathbf{i}_1$ out of the paper ($\odot$): then $\mathbf{j}_1$ points right and $\mathbf{k}_1 = \mathbf{i}_1 \times \mathbf{j}_1$ up. The azimuth θ turns counter-clockwise from $\mathbf{j}_1$: $\mathbf{u} = \cos\theta\,\mathbf{j}_1 + \sin\theta\,\mathbf{k}_1$ (legs of the shaded triangle) and $\mathbf{e}_\theta = d\mathbf{u}/d\theta$. P projects onto this plane at P', at distance $a\sin\phi$ from the axis.

θ turns about the axis OA, so we look along that axis with $\mathbf{i}_1$ out of the paper ($\odot$). With $\mathbf{j}_1$ to the right, $\mathbf{k}_1 = \mathbf{i}_1 \times \mathbf{j}_1$ points up (fingers from $\mathbf{i}_1$, towards the reader, to $\mathbf{j}_1$, right; thumb up). The azimuth starts at $\mathbf{j}_1$ ($\theta = 0$, equatorial plane) and grows counter-clockwise towards $\mathbf{k}_1$ ($\theta = \pi/2$, upper hemisphere), as in the reading of the figure. The unit vector $\mathbf{u}$ of the plane $\mathbf{j}_1\mathbf{k}_1$ in the direction θ is the hypotenuse of the shaded triangle, with the leg $\cos\theta$ along $\mathbf{j}_1$ (adjacent) and $\sin\theta$ along $\mathbf{k}_1$ (opposite):

$$\mathbf{u}(\theta) = \cos\theta\,\mathbf{j}_1 + \sin\theta\,\mathbf{k}_1, \qquad \mathbf{e}_\theta = \frac{d\mathbf{u}}{d\theta} = -\sin\theta\,\mathbf{j}_1 + \cos\theta\,\mathbf{k}_1$$

($\frac{d}{d\theta}\cos\theta = -\sin\theta$, $\frac{d}{d\theta}\sin\theta = \cos\theta$; $|\mathbf{e}_\theta| = 1$ and $\mathbf{u}\cdot\mathbf{e}_\theta = -\cos\theta\sin\theta + \sin\theta\cos\theta = 0$). The minus sign in $\mathbf{e}_\theta$ appears because turning $\mathbf{u}$ a further 90° counter-clockwise takes its tip to the side of $-\mathbf{j}_1$.

The polar angle ϕ: the meridian plane of P

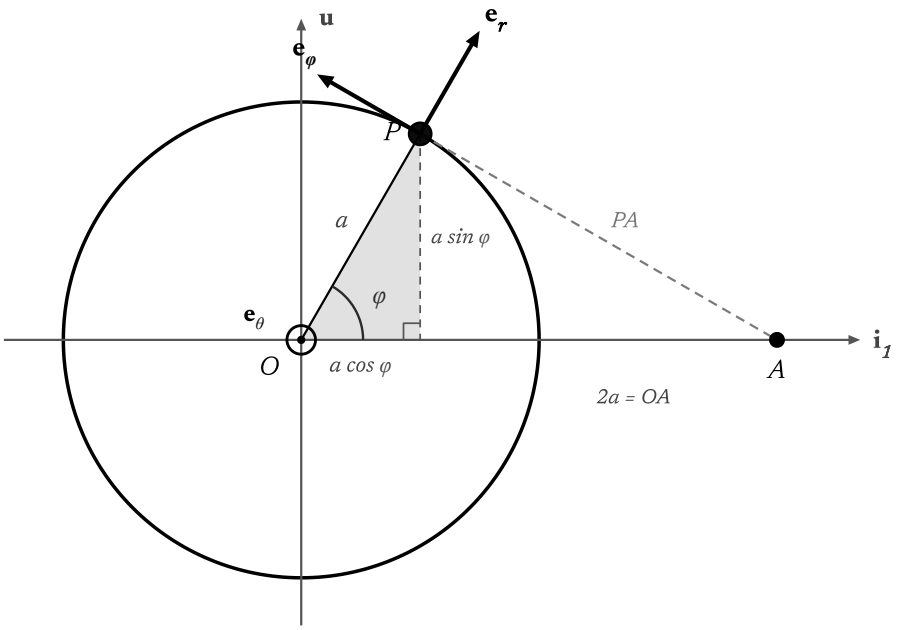


Figure 3. Meridian plane of P (it contains $\mathbf{i}_1$ and $\mathbf{u}$; $\mathbf{e}_\theta = \mathbf{i}_1 \times \mathbf{u}$ is out of the paper). ϕ is measured at O from OA: in the shaded triangle the hypotenuse is a, the leg adjacent to ϕ is $a\cos\phi$ along $\mathbf{i}_1$ and the opposite one $a\sin\phi$ along $\mathbf{u}$. $\mathbf{e}_r, \mathbf{e}_\phi$ in orange; the attraction acts along PA.

P lies in the plane that contains the axis OA and the direction $\mathbf{u}$ (the meridian plane of P); it is drawn with $\mathbf{i}_1$ to the right and $\mathbf{u}$ up, so $\mathbf{i}_1 \times \mathbf{u} = \cos\theta\,\mathbf{k}_1 - \sin\theta\,\mathbf{j}_1 = \mathbf{e}_\theta$ points out of the paper. In the shaded right triangle the hypotenuse is $|OP| = a$, the leg adjacent to ϕ is $a\cos\phi$ along $\mathbf{i}_1$ and the leg opposite to ϕ is $a\sin\phi$ along $\mathbf{u}$ (the distance OP' of the axis view). Hence

$$\mathbf{r}_1^P = \overrightarrow{OP} = a\cos\phi\,\mathbf{i}_1 + a\sin\phi\,\mathbf{u} = a\cos\phi\,\mathbf{i}_1 + a\sin\phi\cos\theta\,\mathbf{j}_1 + a\sin\phi\sin\theta\,\mathbf{k}_1 = a\,\mathbf{e}_r$$

Relation with the standard spherical coordinates. The usual spherical coordinates take the polar axis along $\mathbf{k}$ and the azimuth from $\mathbf{i}$ towards $\mathbf{j}$: $\mathbf{e}_r = \sin\phi\cos\theta\,\mathbf{i} + \sin\phi\sin\theta\,\mathbf{j} + \cos\phi\,\mathbf{k}$. Here the roles are permuted cyclically, $(\mathbf{i},\mathbf{j},\mathbf{k}) \to (\mathbf{j}_1,\mathbf{k}_1,\mathbf{i}_1)$, which keeps the triad right-handed ($\mathbf{j}_1 \times \mathbf{k}_1 = \mathbf{i}_1$); substituting gives exactly the $\mathbf{e}_r$ above. The polar axis is OA instead of the vertical because then $|AP|$ depends on ϕ only.

Spherical basis by differentiation. The radial vector is $\mathbf{e}_r = \cos\phi\,\mathbf{i}_1 + \sin\phi\,\mathbf{u}$. Its derivative with respect to ϕ ($\frac{d}{d\phi}\cos\phi = -\sin\phi$, $\frac{d}{d\phi}\sin\phi = \cos\phi$) already has unit length, $\sqrt{\sin^2\phi + \cos^2\phi} = 1$; with respect to θ, $\frac{\partial \mathbf{e}_r}{\partial\theta} = \sin\phi\,\frac{d\mathbf{u}}{d\theta} = \sin\phi\,\mathbf{e}_\theta$, whose length is $\sin\phi$:

$$\mathbf{e}_\phi = \frac{\partial \mathbf{e}_r}{\partial \phi} = -\sin\phi\,\mathbf{i}_1 + \cos\phi\cos\theta\,\mathbf{j}_1 + \cos\phi\sin\theta\,\mathbf{k}_1, \qquad \mathbf{e}_\theta = \frac{1}{\sin\phi}\frac{\partial \mathbf{e}_r}{\partial\theta} = -\sin\theta\,\mathbf{j}_1 + \cos\theta\,\mathbf{k}_1$$

with $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ given above in terms of $\mathcal{B}_0$. The triad is orthonormal and right-handed: $\mathbf{e}_r \times \mathbf{e}_\phi = \left(\cos^2\phi + \sin^2\phi\right)\mathbf{i}_1 \times \mathbf{u} = \cos\theta\,\mathbf{k}_1 - \sin\theta\,\mathbf{j}_1 = \mathbf{e}_\theta$, hence also $\mathbf{e}_\phi \times \mathbf{e}_\theta = \mathbf{e}_r$, $\mathbf{e}_\theta \times \mathbf{e}_r = \mathbf{e}_\phi$. Inverse relations used below (read from the meridian triangle): $\mathbf{u} = \sin\phi\,\mathbf{e}_r + \cos\phi\,\mathbf{e}_\phi$, so

$$\mathbf{i}_1 = \cos\phi\,\mathbf{e}_r - \sin\phi\,\mathbf{e}_\phi, \qquad \mathbf{k}_1 = \sin\theta\,\mathbf{u} + \cos\theta\,\mathbf{e}_\theta = \sin\phi\sin\theta\,\mathbf{e}_r + \cos\phi\sin\theta\,\mathbf{e}_\phi + \cos\theta\,\mathbf{e}_\theta$$

(the second from the axis-view triangle, $\mathbf{k}_1 = \sin\theta\,\mathbf{u} + \cos\theta\,\mathbf{e}_\theta$).

Kinematics

(a) Position, velocity and acceleration in S_1

Pedagogical Justification: $\mathbf{r}_1^P = a\,\mathbf{e}_r$ (Notes Eq. 3.16 with the origin O common to both frames). The relative derivative keeps $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ fixed, so the spherical vectors change only through $\phi(t)$ and $\theta(t)$ (chain rule).

Time derivatives of the basis in S_1:

$$\left.\frac{d\mathbf{e}_r}{dt}\right|_1 = -\sin\phi\,\dot\phi\,\mathbf{i}_1 + \cos\phi\,\dot\phi\,\mathbf{u} + \sin\phi\,\frac{d\mathbf{u}}{d\theta}\dot\theta = \dot\phi\,\mathbf{e}_\phi + \sin\phi\,\dot\theta\,\mathbf{e}_\theta$$

$$\left.\frac{d\mathbf{e}_\phi}{dt}\right|_1 = -\cos\phi\,\dot\phi\,\mathbf{i}_1 - \sin\phi\,\dot\phi\,\mathbf{u} + \cos\phi\,\dot\theta\,\mathbf{e}_\theta = -\dot\phi\,\mathbf{e}_r + \cos\phi\,\dot\theta\,\mathbf{e}_\theta$$

$$\left.\frac{d\mathbf{e}_\theta}{dt}\right|_1 = \dot\theta\,(-\cos\theta\,\mathbf{j}_1 - \sin\theta\,\mathbf{k}_1) = -\dot\theta\,\mathbf{u} = -\sin\phi\,\dot\theta\,\mathbf{e}_r - \cos\phi\,\dot\theta\,\mathbf{e}_\phi$$

Position and velocity:

$$\mathbf{r}_1^P = a\,\mathbf{e}_r, \qquad \mathbf{v}_1^P = a\left.\frac{d\mathbf{e}_r}{dt}\right|_1 = a\dot\phi\,\mathbf{e}_\phi + a\sin\phi\,\dot\theta\,\mathbf{e}_\theta$$

Acceleration, by the product rule on each term, $\frac{d}{dt}(a\dot\phi\,\mathbf{e}_\phi) = a\ddot\phi\,\mathbf{e}_\phi + a\dot\phi\left(-\dot\phi\,\mathbf{e}_r + \cos\phi\,\dot\theta\,\mathbf{e}_\theta\right)$ and $\frac{d}{dt}(a\sin\phi\,\dot\theta\,\mathbf{e}_\theta) = a\left(\cos\phi\,\dot\phi\dot\theta + \sin\phi\,\ddot\theta\right)\mathbf{e}_\theta + a\sin\phi\,\dot\theta\left(-\sin\phi\,\dot\theta\,\mathbf{e}_r - \cos\phi\,\dot\theta\,\mathbf{e}_\phi\right)$; collecting:

$$\mathbf{a}_1^P = -a\left(\dot\phi^2 + \sin^2\phi\,\dot\theta^2\right)\mathbf{e}_r + a\left(\ddot\phi - \sin\phi\cos\phi\,\dot\theta^2\right)\mathbf{e}_\phi + a\left(\sin\phi\,\ddot\theta + 2\cos\phi\,\dot\phi\dot\theta\right)\mathbf{e}_\theta$$

Forces and Equations of Motion

(b) Forces on the particle

Pedagogical Justification: in S_1, $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eq. 4.63). The actual forces are the attraction and the reaction; the inertia forces (Notes Eq. 4.62) reduce to the centrifugal and Coriolis forces because $\mathbf{a}_0^O = \mathbf{0}$ and $\boldsymbol{\alpha}_{10} = \mathbf{0}$.

Attraction (along the dashed segment PA of the meridian figure): $\mathbf{F}_k = k\left(\mathbf{r}_1^A - \mathbf{r}_1^P\right) = k\,(2a\,\mathbf{i}_1 - a\,\mathbf{e}_r)$; with $\mathbf{i}_1 = \cos\phi\,\mathbf{e}_r - \sin\phi\,\mathbf{e}_\phi$,

$$\mathbf{F}_k = ka\,(2\cos\phi - 1)\,\mathbf{e}_r - 2ka\sin\phi\,\mathbf{e}_\phi$$

Reaction: $\mathbf{N} = N\,\mathbf{e}_r$.

Centrifugal force. With the identity $\mathbf{a}\times(\mathbf{b}\times\mathbf{c}) = (\mathbf{a}\cdot\mathbf{c})\mathbf{b} - (\mathbf{a}\cdot\mathbf{b})\mathbf{c}$ (Slide 30): $\boldsymbol{\omega}_{10}\times\left(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P\right) = \Omega^2 z_1\,\mathbf{k}_1 - \Omega^2\mathbf{r}_1^P$, where $z_1 = \mathbf{r}_1^P\cdot\mathbf{k}_1 = a\sin\phi\sin\theta$. Hence $\mathbf{F}_C = m\Omega^2\left(\mathbf{r}_1^P - z_1\mathbf{k}_1\right)$, and with $\mathbf{k}_1$ in the spherical basis:

$$\mathbf{F}_C = m\Omega^2 a\left[\left(1-\sin^2\phi\sin^2\theta\right)\mathbf{e}_r - \sin^2\theta\sin\phi\cos\phi\,\mathbf{e}_\phi - \sin\phi\sin\theta\cos\theta\,\mathbf{e}_\theta\right]$$

Coriolis force. Using $\mathbf{k}_1 = \sin\phi\sin\theta\,\mathbf{e}_r + \cos\phi\sin\theta\,\mathbf{e}_\phi + \cos\theta\,\mathbf{e}_\theta$ and the cyclic products of the right-handed triad:

$$\mathbf{k}_1\times\mathbf{e}_\phi = \sin\phi\sin\theta\,\mathbf{e}_r\times\mathbf{e}_\phi + \cos\theta\,\mathbf{e}_\theta\times\mathbf{e}_\phi = \sin\phi\sin\theta\,\mathbf{e}_\theta - \cos\theta\,\mathbf{e}_r$$

$$\mathbf{k}_1\times\mathbf{e}_\theta = \sin\phi\sin\theta\,\mathbf{e}_r\times\mathbf{e}_\theta + \cos\phi\sin\theta\,\mathbf{e}_\phi\times\mathbf{e}_\theta = -\sin\phi\sin\theta\,\mathbf{e}_\phi + \cos\phi\sin\theta\,\mathbf{e}_r$$

$$\mathbf{F}_{COR} = -2m\Omega\,\mathbf{k}_1\times\left(a\dot\phi\,\mathbf{e}_\phi + a\sin\phi\,\dot\theta\,\mathbf{e}_\theta\right) = 2m\Omega a\left[\left(\cos\theta\,\dot\phi - \sin\theta\sin\phi\cos\phi\,\dot\theta\right)\mathbf{e}_r + \sin\theta\sin^2\phi\,\dot\theta\,\mathbf{e}_\phi - \sin\theta\sin\phi\,\dot\phi\,\mathbf{e}_\theta\right]$$

(b, continued) Vector equation and scalar equations

Vector equation of motion in S_1:

$$m\mathbf{a}_1^P = k\left(\mathbf{r}_1^A - \mathbf{r}_1^P\right) + N\,\mathbf{e}_r - m\boldsymbol{\omega}_{10}\times\left(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P\right) - 2m\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P$$

Scalar equations free of N: project on $\mathbf{e}_\phi$ and $\mathbf{e}_\theta$, which are perpendicular to $\mathbf{N}$:

$$\boxed{ma\left(\ddot{\phi} - \sin\phi\cos\phi\,\dot{\theta}^2\right) = -2ka\sin\phi - m\Omega^2 a\sin^2\theta\sin\phi\cos\phi + 2m\Omega a\sin\theta\sin^2\phi\,\dot{\theta}}$$

$$\boxed{ma\left(\sin\phi\,\ddot{\theta} + 2\cos\phi\,\dot{\phi}\dot{\theta}\right) = -m\Omega^2 a\sin\phi\sin\theta\cos\theta - 2m\Omega a\sin\theta\sin\phi\,\dot{\phi}}$$

Projection on $\mathbf{e}_r$ (it contains N):

$$-ma\left(\dot{\phi}^2 + \sin^2\phi\,\dot{\theta}^2\right) = N + ka\left(2\cos\phi - 1\right) + m\Omega^2 a\left(1 - \sin^2\phi\sin^2\theta\right) + 2m\Omega a\left(\cos\theta\,\dot{\phi} - \sin\theta\sin\phi\cos\phi\,\dot{\theta}\right)$$

Answers and Checks

(a) Position, velocity and acceleration

$\mathbf{r}_1^P = a\,\mathbf{e}_r$, $\mathbf{v}_1^P = a\dot{\phi}\,\mathbf{e}_\phi + a\sin\phi\,\dot{\theta}\,\mathbf{e}_\theta$, $\mathbf{a}_1^P = -a\left(\dot{\phi}^2 + \sin^2\phi\,\dot{\theta}^2\right)\mathbf{e}_r + a\left(\ddot{\phi} - \sin\phi\cos\phi\,\dot{\theta}^2\right)\mathbf{e}_\phi + a\left(\sin\phi\,\ddot{\theta} + 2\cos\phi\,\dot{\phi}\dot{\theta}\right)\mathbf{e}_\theta$, with the unit vectors defined above in terms of $\mathcal{B}_1$ and, through $[{}_0R_1]$, of $\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0$.

(b) Equations of motion

The forces are $\mathbf{F}_k$, $\mathbf{N}$, $\mathbf{F}_C$ and $\mathbf{F}_{COR}$ above; the vector equation and the two boxed scalar equations for ϕ and θ are the answer.

(c) Conservation of mechanical energy in S_1

> **Pedagogical Justification:** yes, E_1^P is conserved. The sphere is fixed in S_1 and smooth, so $\mathbf{N}\cdot\mathbf{v}_1^P = 0$ ($\mathbf{v}_1^P$ has no $\mathbf{e}_r$ component); $\mathbf{F}_{COR}\cdot\mathbf{v}_1^P = 0$ (Slide 31); the Euler force and $-m\mathbf{a}_0^O$ are zero; the attraction towards the point A, fixed in S_1, is conservative with $V_k = \frac{1}{2}k\left|\mathbf{r}_1^P - \mathbf{r}_1^A\right|^2$; and the centrifugal force has $V_C = -\frac{1}{2}m\Omega^2 r_\perp^2$ because $\boldsymbol{\omega}_{10}$ is constant (Slide 30).

Potentials on the sphere:

$$\left|\mathbf{r}_1^P - \mathbf{r}_1^A\right|^2 = a^2 - 4a^2\,\mathbf{e}_r\cdot\mathbf{i}_1 + 4a^2 = 5a^2 - 4a^2\cos\phi \;\Rightarrow\; V_k = \frac{5}{2}ka^2 - 2ka^2\cos\phi$$

$$r_\perp^2 = x_1^2 + y_1^2 = a^2 - z_1^2 = a^2 - a^2\sin^2\phi\sin^2\theta \;\Rightarrow\; V_C = -\frac{1}{2}m\Omega^2a^2 + \frac{1}{2}m\Omega^2a^2\sin^2\phi\sin^2\theta$$

Dropping the constants, with $T_1^P = \frac{1}{2}m\left|\mathbf{v}_1^P\right|^2 = \frac{1}{2}ma^2\left(\dot\phi^2 + \sin^2\phi\,\dot\theta^2\right)$:

$$\boxed{E_1^P = \frac{1}{2}ma^2\left(\dot\phi^2 + \sin^2\phi\,\dot\theta^2\right) - 2ka^2\cos\phi + \frac{1}{2}m\Omega^2a^2\sin^2\phi\sin^2\theta = \text{const}}$$

Check. Differentiating with the chain and product rules ($\frac{d}{dt}\sin^2\phi = 2\sin\phi\cos\phi\,\dot\phi$, $\frac{d}{dt}\sin^2\theta = 2\sin\theta\cos\theta\,\dot\theta$) and grouping:

$$\frac{dE_1^P}{dt} = a\dot\phi\left[ma\left(\ddot\phi - \sin\phi\cos\phi\,\dot\theta^2\right) + 2ka\sin\phi + m\Omega^2a\sin^2\theta\sin\phi\cos\phi\right] + a\sin\phi\,\dot\theta\left[ma\left(\sin\phi\,\ddot\theta + 2\cos\phi\,\dot\phi\dot\theta\right) + m\Omega^2a\sin\phi\sin\theta\cos\theta\right]$$

By the equations of (b) the brackets equal $2m\Omega a\sin\theta\sin^2\phi\,\dot\theta$ and $-2m\Omega a\sin\theta\sin\phi\,\dot\phi$, so

$$\frac{dE_1^P}{dt} = 2m\Omega a^2\sin\theta\sin^2\phi\,\dot\phi\dot\theta - 2m\Omega a^2\sin\theta\sin^2\phi\,\dot\phi\dot\theta = 0$$

which is exactly $\mathbf{F}_{COR}\cdot\mathbf{v}_1^P = 0$. The constant is fixed by initial conditions (not given).

(d) Equilibrium configurations

> **Pedagogical Justification:** a relative equilibrium has $\dot\phi = \dot\theta = \ddot\phi = \ddot\theta = 0$, so the right-hand sides of the equations of (b) must vanish (the Coriolis terms disappear). The points $\phi = 0, \pi$ are the poles of the coordinates (θ undefined there) and must be examined separately; the symbolic check confirms the list below with Cartesian coordinates and a Lagrange multiplier.

With $\Lambda = m\Omega^2/k$, the two conditions are

$$\sin\phi\left(2k + m\Omega^2\sin^2\theta\cos\phi\right) = 0 \iff \sin\phi\left(2 + \Lambda\sin^2\theta\cos\phi\right) = 0, \qquad \sin\phi\sin\theta\cos\theta = 0$$

- $\sin\phi = 0$**:** both are satisfied. **E1:** $\phi = 0$, $\mathbf{r}_1^P = a\,\mathbf{i}_1$, the point of the sphere nearest to A. **E2:** $\phi = \pi$, $\mathbf{r}_1^P = -a\,\mathbf{i}_1$, the point farthest from A. They exist for every Λ: at both, the attraction and the centrifugal force are radial (along $\pm\mathbf{i}_1$) and the reaction balances them.
- $\sin\phi \neq 0$**:** the second condition needs $\sin\theta = 0$ or $\cos\theta = 0$. If $\sin\theta = 0$, the first becomes $2 = 0$: impossible. If $\cos\theta = 0$ ($\theta = \pm\pi/2$, $\sin^2\theta = 1$), the first gives

$$\cos\phi^* = -\frac{2k}{m\Omega^2} = -\frac{2}{\Lambda}$$

which has a solution with $\sin\phi^* \neq 0$ only if $\Lambda > 2$. **E3, E4** ($\theta = \pm\pi/2$):

$$\mathbf{r}_1^P = -\frac{2a}{\Lambda}\,\mathbf{i}_1 \pm a\sqrt{1 - \frac{4}{\Lambda^2}}\,\mathbf{k}_1$$

They lie in the meridian plane $\mathbf{i}_1\mathbf{k}_1$ that contains A, on the hemisphere opposite to A, symmetric above and below the equator.

Discussion in terms of $m\Omega^2/k$:

- $\Lambda = m\Omega^2/k \leq 2$: only E1 and E2.
- $\Lambda > 2$: E1, E2, E3 and E4. As $\Lambda \to 2^+$, $\cos\phi^* \to -1$ and E3, E4 merge into E2 (a pitchfork bifurcation); as $\Lambda \to \infty$, $\cos\phi^* \to 0$ and they tend to the poles $\pm a\,\mathbf{k}_1$.

Extension (stability, beyond what is asked). Near E1, $V_1 \approx \text{const} + k\left(y_1^2 + z_1^2\right) + \frac{1}{2}m\Omega^2 z_1^2$, a strict minimum, so E1 is stable for every Λ (energy argument). Near E2, $V_1 \approx \text{const} - ky_1^2 + \left(\frac{1}{2}m\Omega^2 - k\right)z_1^2$; along $\mathbf{j}_1$ the linearised motion is $m\ddot{y}_1 = 2ky_1$ (no Coriolis coupling there, because $\mathbf{k}_1 \times \mathbf{v}_1^P$ is normal to the sphere at E2), so E2 is always unstable with growth rate $\sqrt{2k/m}$. E3 and E4 are maxima of V_1; the linearisation including the Coriolis terms (computed numerically for several values $\Lambda > 2$) has eigenvalues with positive real part, so they are unstable as well.

(e) Reaction of the sphere

> **Pedagogical Justification:** the projection of the vector equation on $\mathbf{e}_r$ (written at the end of the previous section) contains N and only known functions once $\phi(t), \theta(t)$ are known.

$$\boxed{\mathbf{N} = -\left[ma\left(\dot{\phi}^2 + \sin^2\phi\,\dot{\theta}^2\right) + ka\left(2\cos\phi - 1\right) + m\Omega^2 a\left(1 - \sin^2\phi\sin^2\theta\right) + 2m\Omega a\left(\cos\theta\,\dot{\phi} - \sin\theta\sin\phi\cos\phi\,\dot{\theta}\right)\right]\mathbf{e}_r}$$

with $\mathbf{e}_r = \cos\phi\,\mathbf{i}_1 + \sin\phi\cos\theta\,\mathbf{j}_1 + \sin\phi\sin\theta\,\mathbf{k}_1$ and $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in terms of $\mathcal{B}_0$ from the change-of-basis section.

Checks

- **Competition of two effects:** the attraction pulls P towards the point nearest A (E1), and its tangential part has stiffness $\sim k$; the centrifugal force pushes P away from the vertical axis with stiffness $\sim m\Omega^2$. The ratio $\Lambda = m\Omega^2/k$ decides whether, on the far side of the sphere, the centrifugal push towards the equator can be balanced by the attraction: only for $\Lambda > 2$ do the extra equilibria E3, E4 appear.
- **Reaction at E1:** with $\phi = 0$ and no motion, $N = -ka - m\Omega^2 a$: the sphere pulls P inwards, since both the attraction (towards A, outside the sphere) and the centrifugal force push P outwards. The bilateral contact assumed among the hypotheses is needed for that.
- **Limit $\Omega \to 0$:** the inertia forces vanish, $\Lambda \to 0$, and only E1 (stable) and E2 (unstable) remain, as for a fixed sphere with a fixed attracting point.
- **Orthogonality:** $\mathbf{e}_r \cdot \mathbf{e}_\phi = -\cos\phi\sin\phi + \sin\phi\cos\phi(\cos^2\theta + \sin^2\theta) = 0$, $\mathbf{e}_r \cdot \mathbf{e}_\theta = \sin\phi(-\cos\theta\sin\theta + \sin\theta\cos\theta) = 0$, $\mathbf{e}_\phi \cdot \mathbf{e}_\theta = \cos\phi(-\cos\theta\sin\theta + \sin\theta\cos\theta) = 0$.
- **Dimensions:** in the $\mathbf{e}_\phi$ equation, $[ma\ddot{\phi}] = \mathrm{kg\,m\,s^{-2}} = \mathrm{N}$, $[ka] = \mathrm{N}$, $[m\Omega^2 a] = \mathrm{N}$, $[m\Omega a\dot{\theta}] = \mathrm{N}$; $[E_1^P] = [ka^2] = [m\Omega^2 a^2] = \mathrm{J}$; $\Lambda = \mathrm{kg\,s^{-2}/(N/m)}$ is dimensionless.
- **Symbolic check (sympy):** $\mathbf{a}_0^P$ from differentiating $[{}_0R_1]\,\mathbf{r}_1^P$ twice equals the Notes Eq. 3.27 decomposition; the force components, both equations of motion, the reaction and $dE_1^P/dt = 0$ are confirmed; the equilibria found with Cartesian coordinates and a Lagrange multiplier are exactly E1 to E4.
- **Check against the official key:** none for this problem (Problems.pdf, p. 76 lists no answer for 49), so the results rest on the independent checks above.

Source: Problems.pdf (PDF page 34); no official key